

Practice by Standard: Number Sense

Standard Set 1.0: Students compute with very large and very small numbers, positive integers, decimals, and fractions and understand the relationship between decimals, fractions, and percents. They understand the relative magnitudes of numbers.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1 What is 4,368.375 rounded to the nearest ten? (5NS1.1)

A 4,368
B 4,370
C 4,368.4
D 4,368.38

- 2 What is 60% of 240? (5NS1.2)

F 125
G 140
H 144
J 166

- 3 $2^5 =$ (5NS1.3)

A 5×5
B $5 + 5$
C $2 + 2 + 2 + 2 + 2$
D $2 \times 2 \times 2 \times 2 \times 2$

QUICK Review

STRATEGY Be sure you round to the appropriate place.

What digit is in the ones place? What digit is in the tens place?

For more help with rounding decimals, see page 142.

READING HINT Percent means *per hundred*.

You can answer this question by finding $\frac{3}{5}$ of 240.

For more help with percent of a number, see page 492.

STRATEGY Think of a power as repeated multiplication.

How many times is 2 being multiplied by itself?

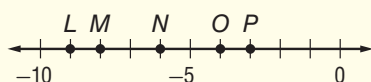
For more help with powers, see page 22.

QUICK Practice

- 4** What is the prime factorization of 24? (5NS1.4)

F $2^3 \times 3$
G $2^2 \times 3^2$
H $3^2 \times 4$
J $3^3 \times 4^2$

- 5** Which letter on the number line best identifies the location of -6 ? (5NS1.5)



A L **C** N
B M **D** O

- 6** What is $\frac{3}{4}$ written as a percent? (5NS1.2)

F 0.75%
G 34%
H 65%
J 75%

- 7** Salvador had a basket of 140 blocks. If 35% of the blocks are red, how many blocks are *not* red? (5NS1.2)

A 49
B 56
C 91
D 94

QUICK Review

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

Multiply the numbers in the remaining choices.

For more help with prime factorization, see page 23.

STRATEGY Write the missing numbers on the number line.

Find the point that corresponds to -6 on the number line.

For more help with locating integers on number lines, see page 114.

READING HINT Percent means *per hundred*. What part of 100 is equal to three-fourths?

You can answer this question by solving

$$\frac{3}{4} = \frac{n}{100}$$

For more help with writing fractions as percents, see page 467.

READING HINT When reading a question, look for key words such as *not* or *both*.

To find the percent of blocks that are not red, subtract 35% from 100%.

For more help with percent of a number, see page 492.

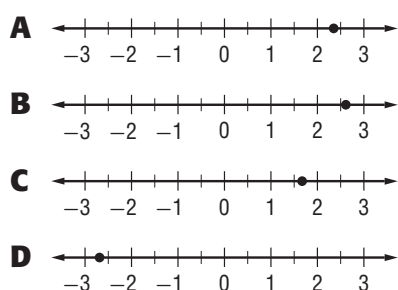
Practice on Your Own

Standard Set 1.0

- 8** What is the prime factorization of 20? (5NS1.4)

F $2^2 \times 3^2$ **H** $2^2 \times 5$
G $2^3 \times 5$ **J** $2^3 \times 3$

- 9** Which point on the number line best represents 2.65? (5NS1.5)



- 10** California is the third largest state in the United States with an area of 163,707 square miles. What is this value rounded to the nearest thousand square miles? (5NS1.1)

F 164,000
G 163,710
H 163,700
J 163,000

- 11** $3^4 =$ (5NS1.3)

A $3 \times 3 \times 3 \times 3$
B $3 + 3 + 3 + 3$
C $4 \times 4 \times 4$
D $4 + 4 + 4$

- 12** What is 7,080.362 rounded to the nearest ten? (5NS1.1)

F 7,100 **H** 7,080.36
G 7,080 **J** 7,080.4

- 13** What is $\frac{1}{8}$ written as a percent? (5NS1.2)

A 0.125% **C** 12.5%
B 1.25% **D** 125%

- 14** An electronics store sold 150 computers last month. If 105 of the computers sold came with flat panel monitors, what percent of the computers came with flat panel monitors? (5NS1.2)

F 80%
G 70%
H 60%
J 45%

- 15** What is the decimal 0.4 written as a fraction? (5NS1.2)

A $\frac{1}{5}$
B $\frac{1}{4}$
C $\frac{2}{5}$
D $\frac{3}{4}$

- 16** What is 75% of 120? (5NS1.2)

F 75 **H** 90
G 82 **J** 93

- 17** What is the prime factorization of 90? (5NS1.4)

A $2^2 \times 3 \times 5^2$
B $2^2 \times 3^2 \times 5$
C $2 \times 3^3 \times 5^2$
D $2 \times 3^2 \times 5$

Practice by Standard: Number Sense

Standard Set 2.0: Students perform calculations and solve problems involving addition, subtraction, and simple multiplication and division of fractions and decimals.

DIRECTIONS

Choose the best answer.

QUICK Practice

1 $14.7 \times 3.9 =$ (5NS2.1)

- A** 57.3
- B** 57.33
- C** 59.13
- D** 59.33

2 $46.62 \div 6.3 =$ (5NS2.2)

- F** 0.67
- G** 0.74
- H** 6.7
- J** 7.4

3 Anna bought $1\frac{1}{8}$ pounds of turkey and $3\frac{1}{4}$ pounds of roast beef. How many pounds of meat did she buy? (5NS2.3)

- A** $3\frac{1}{4}$
- B** $3\frac{7}{8}$
- C** $4\frac{3}{8}$
- D** $4\frac{3}{4}$

QUICK Review

STRATEGY Align the decimal points and then multiply the numbers.

Remember that there should be 2 digits to the right of the decimal point in the product.

For more help with multiplying decimals, see page 290.

STRATEGY Solve a simpler problem.

You can answer this question by multiplying the divisor and dividend by 10. Then perform the division shown below.

$$63 \overline{)466.2}$$

For more help with dividing decimals, see page 303.

STRATEGY Find a common denominator for the fractions.

Write each mixed number using the common denominator of 8. Then add.

For more help with adding mixed numbers, see page 260.

QUICK Practice

4 $\frac{2}{5} \div \frac{4}{9} =$ (5NS2.4)

F $\frac{8}{45}$

G $\frac{4}{5}$

H $\frac{9}{10}$

J $\frac{7}{20}$

5 $\frac{1}{7} \cdot \frac{1}{3} =$ (5NS2.5)

A $\frac{1}{21}$

B $\frac{1}{15}$

C $\frac{1}{10}$

D $\frac{1}{4}$

- 6 Cindy can type 34 words per minute. At this rate, how many words can Cindy type in 8.5 minutes? (5NS2.1)

F 275

G 289

H 296

J 315

7 $5\frac{5}{6} - 2\frac{1}{2} =$ (5NS2.3)

A $3\frac{2}{3}$

B $3\frac{1}{3}$

C $2\frac{2}{3}$

D $2\frac{1}{6}$

QUICK Review

STRATEGY Rewrite the division problem as a multiplication problem by multiplying the dividend by the reciprocal of the divisor.

The division problem $\frac{2}{5} \div \frac{4}{9}$ is equivalent to the multiplication problem $\frac{2}{5} \times \frac{9}{4}$.

For more help with dividing fractions, see page 328.

STRATEGY When multiplying fractions, multiply the numerators. Then multiply the denominators.

What is the product of 7 and 3?

For more help with multiplying fractions, see page 316.

READING HINT The *rate*, or speed, that Cindy can type is 34 words per minute.

You can answer this question by solving $\frac{34}{1} = \frac{x}{8.5}$.

For more help with multiplying decimals, see page 284.

STRATEGY Find a common denominator for the fractions.

Write each mixed number using the common denominator of 6. Then subtract.

For more help with subtracting mixed numbers, see page 260.

Practice on Your Own

Standard Set 2.0

8 $50,594 \div 82 =$ (5NS2.1)

F 576

H 679

G 617

J 713

- 9** Elaine spent $\frac{1}{2}$ hour working on math homework and $\frac{1}{3}$ hour on a reading assignment after school. How much time did she spend doing homework altogether? (5NS2.3)

A $\frac{3}{4}$ hour

B $\frac{5}{6}$ hour

C $\frac{2}{3}$ hour

D $\frac{1}{6}$ hour

- 10** It takes Eduardo $\frac{1}{10}$ hour to ride his bike to Wendy's house and $\frac{1}{5}$ hour to ride from Wendy's house to the mall. How much time does it take him to ride to Wendy's house and then to the mall? (5NS2.3)

F $\frac{3}{10}$ hour

G $\frac{1}{15}$ hour

H $\frac{1}{50}$ hour

J $\frac{2}{5}$ hour

11 $4\frac{5}{8} - 3\frac{1}{2} =$ (5NS2.3)

A $2\frac{1}{4}$

B $2\frac{1}{8}$

C $1\frac{1}{8}$

D $\frac{7}{8}$

- 12** Mrs. Wilson bought 4 pounds of birdseed for her feeder. She saved \$3.16 by using a store coupon. How much did she save per pound of birdseed? (5NS2.1)

F \$1.27

G \$1.09

H \$0.85

J \$0.79

13 27.09
 $\times 0.6$ (5NS2.1)

A 14.574

B 14.804

C 16.164

D 16.254

- 14** A carpenter had a board 4.65 centimeters long. He cut off 1.3 centimeters. How long was the piece of board that was left? (5NS2.1)

F 2.95 centimeters

G 3.15 centimeters

H 3.35 centimeters

J 3.75 centimeters

15 $3\frac{1}{5} + 4\frac{1}{2} =$ (5NS2.3)

A $7\frac{7}{10}$

B $7\frac{2}{3}$

C $7\frac{2}{7}$

D $7\frac{1}{10}$

Practice by Standard: Algebra and Functions

Standard Set 1.0: Students use variables in simple expressions, compute the value of the expression for specific values of the variable, and plot and interpret the results.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1 If $b = 5$, what is the value of $3 \times b - 2$? (5AF1.2)

A 6
B 10
C 11
D 13

- 2 Which situation could be described by the expression $h + 1\frac{2}{3}$? (5AF1.1)

F Lisa worked h hours in her garden yesterday and $1\frac{2}{3}$ fewer hours today.
G Lisa worked $1\frac{2}{3}$ hours in her garden yesterday and h fewer hours today.
H Lisa worked h hours in her garden yesterday and $1\frac{2}{3}$ more hours today.
J Lisa worked h hours in her garden yesterday and $1\frac{2}{3}$ times more hours today.

QUICK Review

STRATEGY Substitute 5 for b and simplify the resulting expression.

This problem can be solved by simplifying the expression below.

$$3 \times 5 - 2$$

For more help with evaluating algebraic expressions, see page 34.

STRATEGY The expression shows $1\frac{2}{3}$ being added to some quantity h . Look for the choice that represents this expression.

Which situation can be represented by a sum?

For more help with using expressions to describe situations, see page 35.

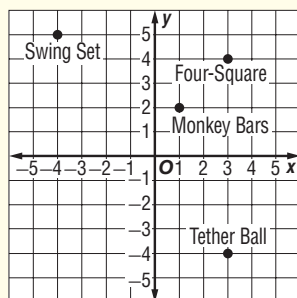
QUICK Practice

- 3 What value for k makes this equation true? (5AF1.3)

$$9 \times 24 = (9 \times 20) + (9 \times k)$$

- A 4
B 7
C 11
D 12

- 4 The map below shows the location of playground equipment.



Which ordered pair *best* names the location of the monkey bars? (5AF1.4)

- F (2, 1) H (-1, 2)
G (1, 2) J (-2, 1)

- 5 Which equation could have been used to create the table shown below? (5AF1.5)

x	-2	-1	1	2
y	-8	-4	4	8

- A $y = x + 4$ C $y = 4x$
B $x = y + 4$ D $x = 4y$

QUICK Review

STRATEGY Use backsolving to find which answer choice is the solution. When you backsolve, you substitute each value for the variable to find which choice makes the left side of the equation equal to the right side.

Substitute each answer choice into the equation for k and see which one results in a true number sentence.

For more help with the Distributive Property, see page 58.

STRATEGY Locate the point on the coordinate grid that represents the monkey bars. Then identify the x -coordinate and y -coordinate of this point.

Remember to list the coordinates of the point as (x, y) .

For more help with naming points on a coordinate grid, see page 378.

STRATEGY Look for a pattern relating x and y .

How can you express this as an equation?

For more help with writing equations for a function, see page 450.

Practice on Your Own

Standard Set 1.0

- 6** If $p = 9$, what is the value of $8p - 5$?

(5AF1.2)

F 67

G 53

H 31

J 12

- 7** Which expression represents the sum of a and 10? (5AF1.2)

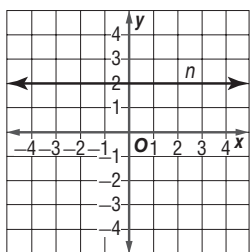
A $a - 10$

B $a + 10$

C $a \div 10$

D $a \times 10$

- 8** Line n is represented by the equation $y = 2$. Which ordered pair is located on line n ? (5AF1.4)



F $(-1, 2)$

G $(0, 0)$

H $(2, 1)$

J $(2, 0)$

- 9** If $t = 48$, what is the value of $9 - t$?

(5AF1.2)

A -57

B -39

C 39

D 57

- 10** Which table represents values x and y such that $y = x + 6$? (5AF1.5)

F

x	y
-2	-8
-1	17

H

x	y
-2	4
-1	5

G

x	y
6	0
2	4

J

x	y
4	-2
3	-3

- 11** Which equation could have been used to create this function table?

(5AF1.5)

x	-8	-5	1	7
y	-10	-7	-1	5

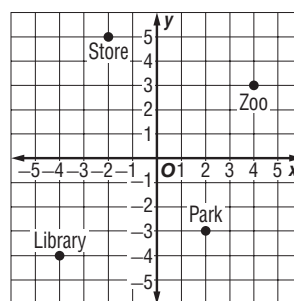
A $y = 2x$

B $y = x + 2$

C $y = x - 2$

D $y = x - 1$

- 12** The map below shows the locations of four different places.



Which place is at the point $(4, 3)$?

(5AF1.4)

F Library

G Park

H Store

J Zoo

Practice by Standard: Measurement and Geometry

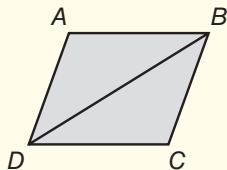
Standard Set 1.0: Students understand and compute the volumes and areas of simple objects.

DIRECTIONS

Choose the best answer.

QUICK Practice

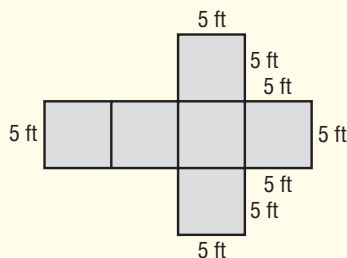
- 1** In the figure below, $ABCD$ is a parallelogram.



If the area of triangle ABD is 15 square centimeters, what is the area of $ABCD$? (5MG1.1)

- A** 15 square centimeters
- B** 30 square centimeters
- C** 45 square centimeters
- D** 60 square centimeters

- 2** What is the surface area of the cube formed by the net below? (5MG1.2)



- F** 25 ft²
- G** 75 ft²
- H** 125 ft²
- J** 150 ft²

QUICK Review

READING HINT Recall that a parallelogram can be split into two congruent triangles.

The area of parallelogram $ABCD$ is equal to twice the area of triangle ABD .

For more help with finding the area of a parallelogram, see page 591.

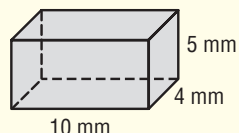
READING HINT *Surface area* in this problem means the sum of the areas of the six faces of the cube.

What is the area of each face of the cube?
How many faces are there?

For more help with finding the surface area of a rectangular prism, see page 618.

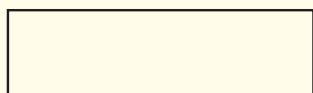
QUICK Practice

- 3 This rectangular prism has a length of 10 millimeters, a height of 5 millimeters, and a width of 4 millimeters. What is the volume? (5MG1.3)



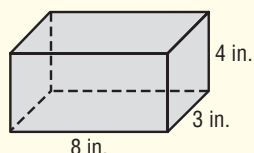
- A 19 cubic millimeters
- B 56 cubic millimeters
- C 200 cubic millimeters
- D 220 cubic millimeters

- 4 Which of the following units would be most appropriate for finding the perimeter of the rectangle shown below? (5MG1.4)



- F cubic inches
- G millimeters
- H square centimeters
- J square feet

- 5 Find the volume of the rectangular prism shown below. (5MG1.3)



- A 15 in^3
- B 48 in^3
- C 96 in^3
- D 192 in^3

QUICK Review

READING HINT The volume of a rectangular prism can be found by using the formula $V = \ell wh$, where ℓ is the length, w is the width, and h is the height.

Replace ℓ with 10, w with 4, and h with 5 in the formula for the volume of a rectangular prism, $V = \ell wh$.

For more help with finding the volume of rectangular prisms, see page 608.

READING HINT Recall the distance around any closed figure is called its *perimeter*.

The perimeter P of a rectangle is the sum of its two lengths and two widths.

For more help with appropriate units of measure for the perimeter of two-dimensional figures, see page 585.

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

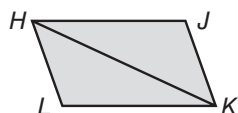
Replace ℓ with 8, w with 3, and h with 4 in the formula for the volume of a rectangular prism, $V = \ell wh$.

For more help with finding the volume of rectangular prisms, see page 608.

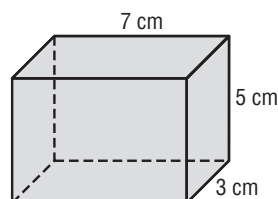
Practice on Your Own

Standard Set 1.0

- 6** In the figure below, $HJKL$ is a parallelogram. If the area of $HJKL$ is 84 square inches, what is the area of triangle LKH ? (5MG1.1)

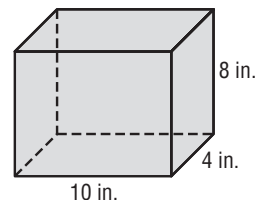


- F** 21 square inches
G 42 square inches
H 84 square inches
J 168 square inches
- 7** The rectangular prism below has a length of 7 centimeters, a height of 5 centimeters, and a width of 3 centimeters. What is its volume? (5MG1.3)

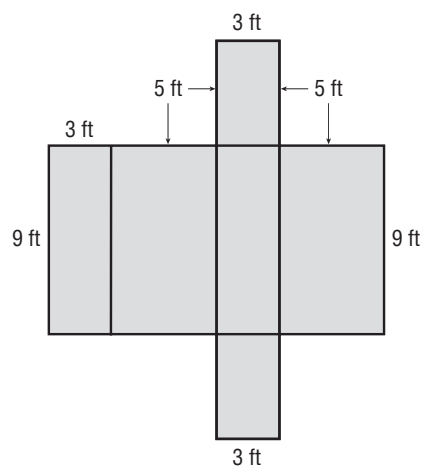


- A** 88 cubic centimeters
B 105 cubic centimeters
C 120 cubic centimeters
D 142 cubic centimeters
- 8** Which of the following units would be most appropriate for finding the surface area of a prism? (5MG1.4)
- F** cubic meters
G cubic inches
H square feet
J yards

- 9** The rectangular prism below has a length of 10 inches, a height of 8 inches, and a width of 4 inches. What is the volume? (5MG1.3)



- A** 22 cubic inches
B 44 cubic inches
C 88 cubic inches
D 320 cubic inches
- 10** What is the surface area of the rectangular prism formed by the net below? (5MG1.2)



- F** 174 square feet
G 161 square feet
H 135 square feet
J 108 square feet

Practice by Standard: Measurement and Geometry

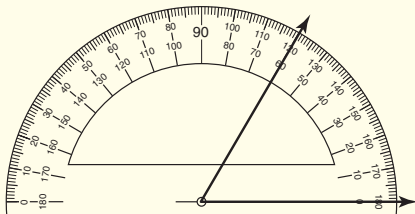
Standard Set 2.0: Students identify, describe, and classify the properties of, and the relationships between, plane and solid geometric figures.

DIRECTIONS

Choose the best answer.

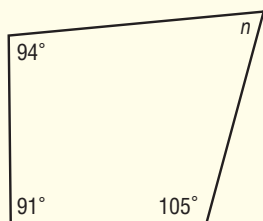
QUICK Practice

- 1 Which is *closest* to the measure of the angle shown below? (5MG2.1)



- A 60°
- B 75°
- C 110°
- D 120°

- 2 What is the measure of angle n in the figure below? (5MG2.2)



- F 160°
- G 135°
- H 70°
- J 55°

QUICK Review

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

The angle measurement should be less than 90°.

For more help with using a protractor to find angle measures, see page 529.

READING HINT The sum of the angles of a quadrilateral is 360°.

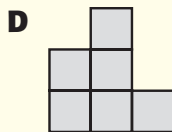
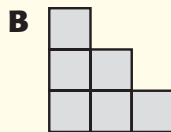
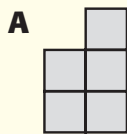
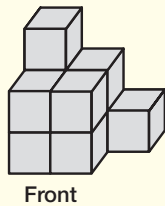
This problem can be solved by solving the equation below.

$$n + 94 + 91 + 105 = 360$$

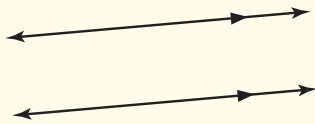
For more help with finding missing angle measures of quadrilaterals, see page 558.

QUICK Practice

- 3** Which of the following shows the front view of the solid below?
(5MG2.3)

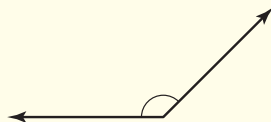


- 4** Which of the following *best* describes the figure below? (5MG2.1)



- F** intersecting lines
G parallel lines
H perpendicular lines
J acute angles

- 5** What is the approximate measure of the angle below in degrees?
(5MG2.1)



- A** 75° **C** 130°
B 100° **D** 175°

QUICK Review

STRATEGY Try to imagine looking at the solid from the front view.

How many blocks would you see in each stack if you were to look at the solid from the front?

For more help with visualizing two-dimensional views of three-dimensional objects, see page 566.

STRATEGY Look at the terms listed. Define the ones that you can and eliminate any choices.

Parallel lines are lines that never intersect.

For more help with identifying parallel lines, see page 541.

READING HINT Recall that an obtuse angle is any angle that measures greater than 90° but less than 180°.

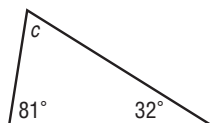
The angle measurement should be greater than 90°.

For more help with estimating angle measures, see page 536.

Practice on Your Own

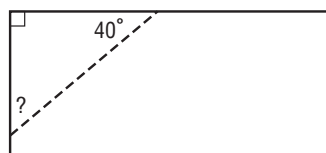
Standard Set 2.0

- 6 What is the measure of angle c in the figure below? (5MG2.2)



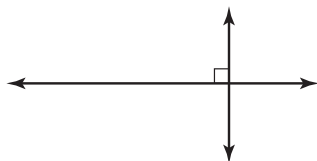
- F 157°
- G 143°
- H 72°
- J 67°

- 7 Kurt made a triangular garden in the corner of his yard. One angle is 40° . What is the measure of the third angle of Kurt's triangle? (5MG2.2)



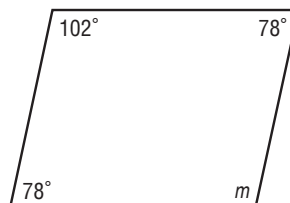
- A 50°
- B 60°
- C 120°
- D 140°

- 8 Which of the following *best* describes the figure below? (5MG2.1)



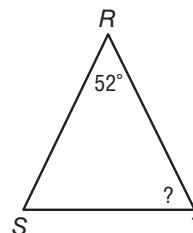
- F obtuse angles
- G parallel lines
- H opposite angles
- J perpendicular lines

- 9 What is the measure of angle m in the figure shown below? (5MG2.2)



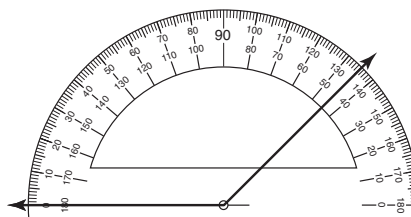
- A 102°
- B 100°
- C 78°
- D 12°

- 10 Triangle RST is isosceles. If the measure of $\angle R$ is 52° and the measures of $\angle S$ and $\angle T$ are equal, what is the measure of $\angle T$? (5MG2.2)



- F 64°
- G 72°
- H 90°
- J 128°

- 11 Which is *closest* to the measure of the angle shown below? (5MG2.1)



- A 45°
- B 55°
- C 135°
- D 145°

Practice by Standard: Statistics, Data Analysis, and Probability

Standard Set 1.0: Students display, analyze, compare, and interpret different data sets, including data sets of different sizes.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Karen's golf scores over her last five rounds are listed below.

75, 79, 80, 77, 76

What is the median of these numbers? (SSDAP1.1)

- A** 75 **C** 77
B 76 **D** 78

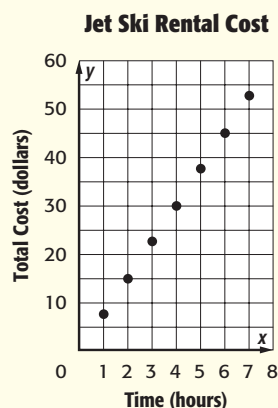
QUICK Review

READING HINT The *median* of a data set is the middle number when the data are arranged in order from least to greatest.

Write the golf scores in order from least to greatest to find the median.

For more help with finding the median of a data set, see page 99.

- 2** The graph shows the cost of renting a jet ski for x hours. If Reggie's total cost was \$30, for how many hours did he rent the jet ski? (SSDAP1.4)



- F** 3 **H** 5
G 4 **J** 6

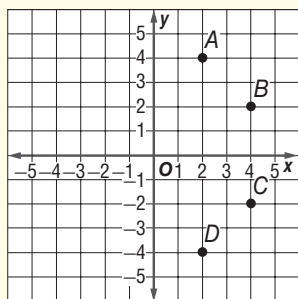
READING HINT An *ordered pair* is a pair of numbers used to locate a point in the coordinate system. The ordered pair is written in this form (x-coordinate, y-coordinate).

Locate the point on the graph that has a y-value of 30. What is the x-value of this point?

For more help with interpreting data on a graph, see page 79.

QUICK Practice

- 3 Which point represents $(2, -4)$ on this graph? (SSDAP1.5)



- A point A
- B point B
- C point C
- D point D

- 4 Which type of graph would be best for displaying the height of a plant over the past several weeks? (SSDAP1.2)

- F bar graph
- G circle graph
- H histogram
- J line graph

- 5 There are seven fourth-graders and thirteen fifth-graders in the school play. What percent of students in the school play are fifth-graders? (SSDAP1.3)

- A 72%
- B 65%
- C 54%
- D 35%

QUICK Review

READING HINT Recall that the x -coordinate corresponds to a number on the x -axis and the y -coordinate corresponds to a number on the y -axis.

Locate the point that is 2 units right of the origin and 4 units below the origin.

For more help with identifying the point corresponding to an ordered pair, see page 378.

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

Are bar graphs, histograms, or circle graphs appropriate for showing how data change over time?

For more help with choosing appropriate types of graphs, see page 106.

READING HINT Percent means *per hundred*.

Solve $\frac{13}{20} = \frac{n}{100}$.

For more help with using percents to represent data, see page 468.

Practice on Your Own

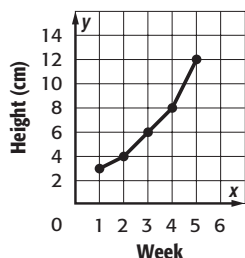
Standard Set 1.0

- 6** What is the mean of the following set of quiz scores? (SSDAP1.1)

8, 9, 7, 6, 10, 9, 7

- F** 7
- G** 7.5
- H** 8
- J** 8.5

- 7** The line graph shows the height of a plant after several weeks. What was the height of the plant during week 4? (SSDAP1.4)



- A** 12 cm
- B** 8 cm
- C** 6 cm
- D** 4 cm

- 8** The table below shows the ages of the players on Aaron's soccer team. What is the mode of the data set? (SSDAP1.1)

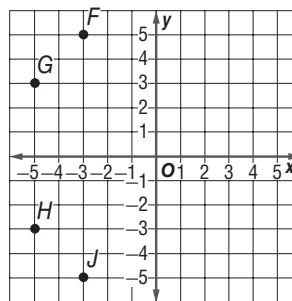
12	12	12	11	9
12	10	9	11	11
10	10	12	12	10

- F** 9
- G** 10
- H** 11
- J** 12

- 9** Which type of graph would be best for comparing the parts of a budget to the whole budget? (SSDAP1.2)

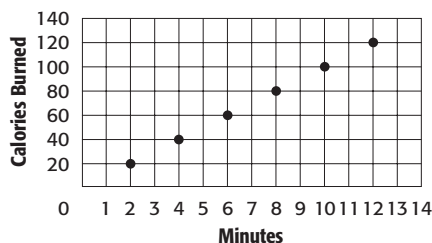
- A** bar graph
- B** circle graph
- C** histogram
- D** line graph

- 10** Which point represents $(-3, 5)$ on this graph? (SSDAP1.5)



- F** point F
- G** point G
- H** point H
- J** point J

- 11** Sam kept track of the number of Calories that he burned per minute while running on the treadmill. How long did it take him to burn 60 Calories? (SSDAP1.5)



- A** 4 minutes
- B** 6 minutes
- C** 8 minutes
- D** 10 minutes

Practice by Standard: Mathematical Reasoning

Standard Set 1.0: Students make decisions about how to approach problems.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Consider the following problem.

Mandy's room is 14 feet long and 10 feet wide with a ceiling that is 8 feet high. Her parents want to have new carpet installed in the room. The carpeting costs \$3.00 per square foot. How much will it cost to have the carpet installed in Mandy's room?

Identify the irrelevant information in the problem. (5MR1.1, 5MG1.4)

- A** the length of Mandy's room
- B** the width of Mandy's room
- C** the height of Mandy's room
- D** the cost of the carpet

- 2** A total of 150 students went on a field trip. Sixty percent of the students brought their own lunches. To mentally find how many students brought their own lunch to the field trip, which of the following could you find first? (5MR1.2, 5NS1.2)

- F** 10% of 60 **H** 150% of 60
- G** 60% of 10 **J** 10% of 150

QUICK Review

STRATEGY Think: What information do you need to solve the problem? Do not be distracted by extra information.

To find the area of Mandy's room, you need to know the length and the width of the room.

For more help with distinguishing relevant from irrelevant information in a problem situation, see page 104.

STRATEGY Sometimes it is helpful to break a complex problem into a simpler one. This can involve doing one step of the problem, using smaller numbers, or rounding.

Solve a simpler problem by finding 10% of the number of students who went on the field trip.

For more help with using mental math to solve a simpler problem, see page 482.

Practice on Your Own

Standard Set 1.0

- 3** The total area of California is 424,000 square kilometers. Of that, about 5% is inland water. Which of the following would you do first to find the estimated amount of California that is inland water?

(5MR1.2, 5NS1.1)

- A** round 424,000
- B** find 95% of 424,000
- C** find 5% of 95
- D** subtract 5% from 100%

- 4** The figures below represent the first three terms of a pattern. If the pattern continues, how many squares will be in Figure 6 of the pattern? (5MR1.1, 5AF1.5)

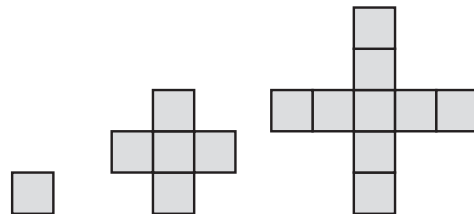


Figure 1

Figure 2

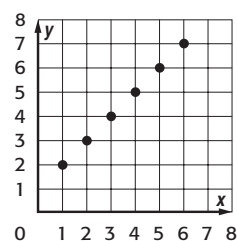
Figure 3

- F** 17
- G** 21
- H** 24
- J** 27

- 5** A total of 29 students are going on a field trip. Each student must pay \$4.65 for transportation and \$5.50 for lunch. About how much money should the teacher collect in all from the students? (5MR1.2, 5NS1.1)

- A** \$225
- B** \$250
- C** \$260
- D** \$300

- 6** Which equation *best* describes the relationship between x and y shown in the graph? (5MR1.1, 5SDAP1.4)



- F** $y = x + 2$
- G** $y = x + 1$
- H** $y = x - 1$
- J** $y = x - 2$

- 7** Consider the following problem.

There are 450 students and 30 teachers at Pine Villa Elementary School. Eighty percent of the students ride the bus to school. How many students do *not* ride the bus to school?

Identify the irrelevant information in the problem. (5MR1.1, 5NS1.2)

- A** the number of students at Pine Villa Elementary School
- B** the percent of students who ride the bus to school
- C** the number of teachers at Pine Villa Elementary School
- D** the percent of students who do not ride the bus to school

- 8** If the pattern continues, what will be the next number? (5MR1.1, 5AF1.5)

3, 9, 15, ?

- F** 18
- G** 21
- H** 27
- J** 45

Practice by Standard: Mathematical Reasoning

Standard Set 2.0: Students use strategies, skills, and concepts in finding solutions.

DIRECTIONS

Choose the best answer.

QUICK Practice

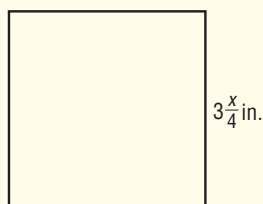
- 1** The table shows the number of points Kris' football team scored last season.

14	21	24	17	14	21	9	34	15	17
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Kris calculated that the mode of the scores was 16. Which of the following shows that his calculation is invalid? **(5MR2.6, 5SDAP1.1)**

- A** The median is 16, not 17.
- B** Most scores are above the mode.
- C** The team never scored 16 points.
- D** The mean is 17, not 16.

- 2** Which expression represents the perimeter in inches of the square shown below? **(5MR2.3, 5MG1.4)**



- F** $3\frac{x}{4} + 3\frac{x}{4}$
- G** $2(3\frac{x}{4}) + 3\frac{x}{4}$
- H** $4(3\frac{x}{4})$
- J** $4(3\frac{x}{4}) \times \frac{1}{2}$

QUICK Review

STRATEGY You do not have to calculate the mode in order to answer this problem.

You are looking for a true statement that shows that 16 cannot be the mode. Read through each answer choice, checking for its validity and how it shows that Kris' calculation of the mode cannot be true.

For more help with the mode of a data set, see page 99.

READING HINT The perimeter of a square can be found by using the formula $P = 4s$.

Replace s with $3\frac{x}{4}$ in the formula for the perimeter of a square.

For more help with the perimeter of a square, see page 585.

Practice on Your Own

Standard Set 2.0

- 3** Which of the following is true concerning the area of the rectangle shown below? (5MR2.5, 5MG1.4)



- A** The exact area is 40 ft^2 .
B The approximate area can be found by multiplying 8 by $5\frac{1}{4}$.
C The exact area can be found by multiplying 8 by 5.
D The approximate area can be found by multiplying 8 by 5.
- 4** Paperback books cost \$5.79 including tax at Book Smart. Juliana buys 4 paperback books. She determined the cost of the 4 books is \$23.16. Which of the following verifies that her solution is reasonable? (5MR2.1, 5NS1.1)
- F** $5.79 \approx 6$, so $6 \times 4 = 24$
G $5.79 \approx 5$, so $5 \times 4 = 20$
H $23.13 \approx 20$, so $20 \div 4 = 5$
J $23.13 \approx 20$, so $20 \div 5 = 4$
- 5** Suppose the population of a city was 2,900. Of that, 40% of the people owned dogs. Which of the following could you evaluate first to mentally calculate the number of people who owned dogs? (5MR2.2, 5NS1.2)
- A** 5% of 2,900 **C** 60% of 40
B 10% of 2,900 **D** 40% of 60

- 6** The equation $25 + b = 40$ describes the number of minutes Rebecca spent cleaning her room over two days. Which of the following is true? (5MR2.4, 5AF1.1)

- F** $25 + 40 = 65$, so $b = 65$
G $40 - 25 = 15$, so $b = 15$
H $25 - 40 = -15$, so $b = -15$
J $25 \times 1.6 = 40$, so $b = 1.6$

- 7** Which expression represents the perimeter in feet of the rectangle shown below? (5MR2.3, 5MG1.4)



- A** $13 + 8$
B $6\frac{1}{2} + 8$
C $6\frac{1}{2} + 4x$
D $13 + \frac{4}{x}$
- 8** For each step that Christian takes, he travels a distance of 3 feet. Let x represent the number of steps that he takes and let y represent the distance traveled. Which of the following equations shows the relationship between x and y ? (5MR2.4, 5AF1.5)
- F** $x = 3y$
G $x = y + 3$
H $y = x + 3$
J $y = 3x$

Practice by Standard: Mathematical Reasoning

Standard Set 3.0: Students move beyond a particular problem by generalizing to other situations.

DIRECTIONS

Choose the best answer.

QUICK Practice

- 1** Luz determined that the next term in the sequence below is 35.

55, 48, 41, 34, 27, ...

Which of the following shows that her solution is *not* reasonable?

(5MR3.1, 5AF1.5)

- A** The terms are increasing.
- B** The terms are decreasing.
- C** The terms are multiples of 5.
- D** The terms are odd numbers.

- 2** Use the table to make a generalization about the angle measures of quadrilateral $ABCD$.

(5MR3.3, 5MG2.2)

$m\angle A$	$m\angle B$	$m\angle C$	$m\angle D$
99°	81°	99°	81°
40°	146°	116°	58°
65°	95°	80°	120°

- F** $m\angle A + m\angle B + m\angle C + m\angle D = 360^\circ$
- G** $m\angle B + m\angle C + m\angle D = 360^\circ$
- H** $m\angle A + m\angle B = 180^\circ$
- J** $m\angle A + m\angle B + m\angle C + m\angle D = 180^\circ$

QUICK Review

STRATEGY Think: Which answers can you eliminate because they are not reasonable?

The pattern in the sequence shows that each term is 7 less than the previous term.

For more help with sequences, see page 442.

READING HINT Recall that the sum of the measures of the angles of a quadrilateral is 360° .

Verify that the sum of each set of angles in the table is 360° . Then write an equation representing the sum of the angles of quadrilateral $ABCD$.

For more help with the sum of the angles of quadrilaterals, see page 558.

Practice on Your Own

Standard Set 3.0

- 3** The difference of $\frac{3}{5}$ and $\frac{1}{5}$ is $\frac{2}{5}$, the difference of $\frac{4}{7}$ and $\frac{2}{7}$ is $\frac{2}{7}$, and the difference of $\frac{7}{11}$ and $\frac{5}{11}$ is $\frac{2}{11}$. Use this pattern to make a generalization about the difference of the fractions $\frac{a}{b}$ and $\frac{c}{b}$. (5MR3.3, 5AF1.1)

A $\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b}$

B $\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b^2}$

C $\frac{a}{b} - \frac{c}{b} = \frac{a - c}{b - b}$

D $\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$

- 4** Ryan wants to save 40% of his paycheck each month. In April, his paycheck was \$320.98. He determines that he should save \$192. Which of the following shows that his solution is *not* reasonable? (5MR3.1, 5NS1.1)

F 40% of \$320.98 $\approx \frac{1}{4}$ of \$320, or \$80.00

G 40% of \$320.98 $\approx \frac{1}{2}$ of \$320, or \$160.00

H 40% of \$320.98 $\approx \frac{2}{5}$ of \$320, or \$128.00

J 40% of \$320.98 $\approx \frac{1}{3}$ of \$320, or \$107.00

- 5** Jessie made 15 key chains in 4 hours. Which of the following is a reasonable calculation for the number of hours it would take her to make 44 key chains? (5MR3.1, 5NS2.5)

A 16 hours **C** 10 hours

B 11 hours **D** 8 hours

- 6** Nemo draws a diagonal that splits a rectangle into two congruent triangles. Using the formula for the area of a rectangle, which of the following would give the area of each triangle? (5MR3.2, 5MG1.4)

F $A = \frac{1}{2}\ell w$

H $A = 2\ell w$

G $A = \ell w^2$

J $A = \ell w + 2\ell w$

- 7** Jean determined that cost of renting a popcorn machine for 9 hours was \$82.

Number of Hours	Cost (\$)
1	32
3	47
5	62
7	77

Which of the following shows that her solution is *not* reasonable? (5MR3.1, 5AF1.1)

A The terms are multiples of 15.

B The terms are decreasing.

C The terms are multiples of 11.

D The terms are increasing by 15.

- 8** The table gives several possible angle measures for triangle XYZ.

$m\angle X$	$m\angle Y$	$m\angle Z$
99°	54°	27°
40°	30°	110°
59°	84°	37°

Use the table to make a generalization about the angle measures of a triangle. (5MR3.3, 5MG2.2)

F $m\angle X + m\angle Y + m\angle Z = 360^\circ$

G $m\angle X + m\angle Y - m\angle Z = 180^\circ$

H $m\angle X + m\angle Z = 90^\circ$

J $m\angle X + m\angle Y + m\angle Z = 180^\circ$